

**General Instrument
Standard**

Project No.: 01000
KUS Doc No.: 01000-IA-SPC-0001
Contr. Doc No.:
Rev.: R00
Page: 2 / 49

CONTENTS

1. GENERAL.....	3
2. CODES, STANDARDS, REGULATIONS AND REFERENCES.....	ERROR! BOOKMARK NOT DEFINED.
2.1. APPLICABLE STANDARDS AND CODES	ERROR! BOOKMARK NOT DEFINED.
2.2. REFERENCE DOCUMENTS.....	ERROR! BOOKMARK NOT DEFINED.
2.3. DEVIATIONS FROM PROJECT AND INTERNATIONAL STANDARDS.....	ERROR! BOOKMARK NOT DEFINED.
3. FACILITIES TO BE PROVIDED WITH HVAC SYSTEM	ERROR! BOOKMARK NOT DEFINED.
4. DESIGN PARAMETER.....	ERROR! BOOKMARK NOT DEFINED.
5. GENERAL DESIGN CRITERIA	ERROR! BOOKMARK NOT DEFINED.
5.1. Equipment Sparing Philosophy	ERROR! BOOKMARK NOT DEFINED.
5.2. Refrigerants	ERROR! BOOKMARK NOT DEFINED.
5.3. Air Recirculation limitations	ERROR! BOOKMARK NOT DEFINED.
5.4. Dedicated Exhaust Air Systems.....	ERROR! BOOKMARK NOT DEFINED.
5.5. Clean Agent I Smoke Exhaust System	ERROR! BOOKMARK NOT DEFINED.
5.6. Smoke Exhaust System.....	ERROR! BOOKMARK NOT DEFINED.
5.7. Escape Route Pressurization Systems.....	ERROR! BOOKMARK NOT DEFINED.
5.8. Electrical Power Supply Requirements.....	ERROR! BOOKMARK NOT DEFINED.
5.9. HVAC System Operations	ERROR! BOOKMARK NOT DEFINED.
5.10. Automatic Change-Over Requirements	ERROR! BOOKMARK NOT DEFINED.
5.11. SEASONAL CHANGE-OVER OF SET POINTS	ERROR! BOOKMARK NOT DEFINED.
5.12. ELECTRICAL HAZARDOUS AREA CLASSIFICATION	ERROR! BOOKMARK NOT DEFINED.
5.13. BLAST PROTECTION	ERROR! BOOKMARK NOT DEFINED.
5.14. FIRE PROTECTION	ERROR! BOOKMARK NOT DEFINED.
5.15. GAS PROTECTION	ERROR! BOOKMARK NOT DEFINED.
5.16. INTERCONNECTIONS WITH OTHER SYSTEMS.....	ERROR! BOOKMARK NOT DEFINED.
5.17. FRESH AIR INTAKES	ERROR! BOOKMARK NOT DEFINED.
5.18. EXHAUST AIR OUTLETS	ERROR! BOOKMARK NOT DEFINED.
5.19. NOISE CRITERIA	ERROR! BOOKMARK NOT DEFINED.
5.20. HVAC SYSTEMS DESCRIPTION	ERROR! BOOKMARK NOT DEFINED.

1. INTRODUCTION

The purpose of this specification document is to provide the basis of design for the instrumentation and the associated installation, of the Karabatan Power Plant Project.

2. ABBREVIATIONS

Abbreviations	Description
C&E	Cause and Effect Diagram
C&I	Control & Instrumentation
CCB	Central Control Building
CCS	Compressor Control System
CENELEC	The European Committee for Electro-technical Standardization
dB	Decibels
DCS	Distributed Control System
ESD	Emergency Shut Down
FGS	Fire and Gas System
FAT	Factory Acceptance Test
FO	Fibre Optic
HMI	Human Machine Interface
I/O	Input and Output
MMCS	Machine Monitoring And Conditioning System
IP	Ingress Protection
IS	Intrinsic Safe
LAN	Local Area Network
PD	Positive Displacement Meter
PA	Public Address System
PCS	Process Control System
RTU	Remote Terminal Unit
SAT	Site Acceptance Test
SCADA	Supervisory Control and Data Acquisition

**General Instrument
Standard**

Project No.: 01000
KUS Doc No.: 01000-IA-SPC-0001
Contr. Doc No.:
Rev.: R00
Page: 4 / 49

SS	Stainless Steel
TUV	Technical Inspection Association of Germany
UPS	Uninterruptible Power Supply

3. REFERENCE DOCUMENTS AND STANDARDS

3.1. Project Documents

To be developed by Contractor

- Control Systems Philosophy And Specification
- Unit, Equipment, Line & Instrument Numbering Procedure
- General Site Information: Meteorological Geological And Utilities Data (01000-EM-SPC-0003)
- Package Unit Instrumentation Standard
- Process Analyser Systems And House Standard
- Instrument Installation Standard
- Pressure Gauge Standard
- Temperature Gauge Standard
- Electronic Transmitter Standard
- Temperature Elements And Thermowell Standard
- Primary Flow Elements Standard
- Control Valve Standard
- On/Off Valve Standard
- Instrument Cable Material Standard
- Instrument Piping Materials Standard
- Rotating Equipment Controls Standard
- Machinery Monitoring Standard
- Units of Measurement Specification

3.2. International Standards

<u>Standard</u>	<u>Title</u>
ANSI B1.20.1	Pipe Threads, General Purpose
ANSI B16.1	Cast iron Pipe Flanges and Flanged Fitting
ANSI B16.10	Face to Face Dimensions for Ferrous Valves

**General Instrument
Standard**

Project No.: 01000
KUS Doc No.: 01000-IA-SPC-0001
Contr. Doc No.:
Rev.: R00
Page: 5 / 49

ANSI B16.5	Pipe Flanges and Flanged Fittings
ANSI B16-34	Hydraulic test for control valves
ANSI B16.47A	Steel Flanges (26" and up)
ANSI B40.1	Gauges and Pressure Indicating Dial Type, Elastic Element
ANSI B46.1	Flange Surface Finish
ANSI FCI.70-2	Control Valve Seat Leakage
ANSI B16.36	Orifice Flanges
ANSI/ISA MC96.1	Temperature Measurement Thermocouples
ANSI/ISA S75.01	Flow equations for sizing control valve
ANSI/ISA S75.03	Face-to-face dimensions for flanged globe style control valve bodies (ANSI Classes 125, 150, 250, 300 and 600)
ANSI/ISA S75.04	Face to face dimensions for flangeless control valves
ANSI/ISA S75.16	Face-to-face dimensions for flanged globe style control valve bodies (ANSI Classes 900, 1500 and 2500)
API521	Guide for pressure relief and depressurising systems
API670	Specification for non-contacting vibration and thrust position monitoring system
API Manual	Petroleum Measurement Standards.
API RP520 Parts 1 & 11	Design and Installation of Pressure Relieving Systems in Refineries.
API RP550 Part I	Installation of Refinery Instruments and Control Systems.
API RP550 Part 2	Process Stream Analysers
API RP551	Process Measurement Instrumentation
API RP553	Refinery control valves
API RP555	Process Analysers
API Std 526	Flanged Steel Safety Relief Valves
API Std 527	Commercial Seat Tightness of Safety Relief Valves with metal-to-metal seats.
API 598	Valve Inspection and testing
ASME I	Power Boilers (Relief Valves)
ASME VIII	Unfired Pressure Vessels
ASME PTC 19.3	Performance Test Code - Temperature Measurements
IEC 1024	Protection of Structures against Lightning

**General Instrument
Standard**

Project No.: 01000
KUS Doc No.: 01000-IA-SPC-0001
Contr. Doc No.:
Rev.: R00
Page: 6 / 49

IEC 60079	Code of Practice for the Selection, Installation and Maintenance of Electrical Apparatus for use in Potentially Explosive Atmospheres.
IEC 60092 375	General Instrumentation, Control and Communication Cables
IEC 60331	Fire Resisting Characteristics of Electrical Cables
IEC 60228	Conductors of insulated cables
IEC 60332	Tests on Electrical Cables under Fire conditions
IEC 60529	Degrees of protection provided by enclosures (IP Code)
IEC 60584	Temperature Measurement Thermocouples.
IEC 60751	Industrial Platinum Resistance Thermometer Elements.
IEC 61000	Electromagnetic Compatibility for Industrial Process Measurement and Control Equipment
IEC 61131	Programmable Controllers
IEC 61508	Functional Safety of Electrical / Electronic / Programmable Electronic Safety Related Equipment
IEC 61511	Functional Safety - Safety instrumented system for the Process industry sector
ISA S5.1	Instrumentation symbols and identification
ISA S5.2	Binary Logic Diagrams for Process Operations.
ISA S5.4	Instrument Loop Diagrams
ISA 75.01	Flow equations for Sizing Control Valves
ISA 75.03	Valve Bodies, Face to Face Dimensions for Integral Flanged Globe Style Control Valve Bodies
ISO 5167	Specification for Square-edges Orifice Plates, Nozzles and Venturi Tubes Inserted in Circular Cross Section Conduits Running Full
ISO 5168	Measurement of fluid flow- procedure for the evaluation of uncertainties
ISA-S84.01	Application of Safety Instrumented System For the Process Industries
NACE - MR – 0175	Standard Material Requirements Sulphide Stress Cracking Resistant Metallic Materials for Oilfield Equipment Corrosion Engineers Handbook
BS 6739	Instrumentation in Process Control Systems: Installation and Design Practice
BS 6755	Testing & inspection of valves

The requirements of national legislation, regulations or codes shall not be affected by technical specifications. They shall take precedence over the international code and standards cited in this document.

In the event of any conflict, the Contractor shall refer the matter to the purchase representative for clarification.

4. ENGINEERING AND DESIGN REQUIREMENTS

4.1. General

Licensors(s) Design Recommendations shall form the basis for the control and instrumentation scope of work, for the Karabatan Power Plant

CONTRACTOR shall supplement the Licensors(s) Design Recommendations by the following to further define the objectives where required:

- Process Control Narratives detailing all control and monitoring requirements including sequencing, calculations, interlocks, Cause & Effect (C&E), etc.
- Safeguarding Narratives detailing all Emergency Shutdown requirements in narrative form and supplemented by Cause & Effect Diagrams.
- Fire and Gas Narratives detailing all fire and gas safeguarding requirements in narrative form and supplemented by Cause & Effect Diagrams.

Fire and gas detection and annunciation shall be detailed on drawings using plot plan layout as background.

All package equipment control, monitoring and safeguarding requirements shall also be incorporated within the above documentation to the same standards to ensure consistency do the overall project.

4.2. Tagging and Identification

Tagging and identification shall be in accordance with EQUIPMENT, LINE & INSTRUMENT NUMBERING PROCEDURE **108184-0000-EM-SPC-0002** and P&ID Legend Detail (to be provided by Contractor).

Control and instrumentation database shall be generated within Intergraph's INtools software package as the basis for control and instrumentation engineering and design within the following modules:

- Instrument Index
- Process Data
- Specification Sheets
- Wiring
- Loop Drawings
- etc

CONTRACTOR shall be required to handover an As-Built INtools database following final acceptance for use by COMPANY.

4.3. Language

English language shall be used in all documents and drawings.

The Russian and/or Kazakh language shall be used in operation manual and human manual interface.

4.4. Units

All units and measurements shall be in metric in accordance with ISO 1000- 1973 " International Systems of Units" SI units of measurement supplemented by the following:

- Inches (for pipe diameters)
- MMscfd
- Bar

Refer to Units of Measurement Specification (to be developed by Contractor).

5. GENERAL REQUIREMENTS

5.1. Environmental Conditions

All instrumentation and associated installation shall be designed and supplied suitable for use in a hot, humid and dusty environment and protected against wind and stormy weather conditions. In addition the instrumentation and associated installation shall be protected from snow and snow storms where the temperature can reach -40 degree Celsius.

5.2. Instrument Protection

The minimum degree of protection for field mounted instrumentation shall be IP65 in accordance with IEC 60529.

All field-mounted instruments shall be tropical-climate-resistant to prevent condensation damage, mould growth and bug/ insect damage.

Sun-shields shall be provided for all electronic instrumentation in direct sunlight.

Suitable protection shall be provided during the cold seasons such as heated instrument housing. Installation shall also be provided with winterization as required.

5.3. Hazardous Area Classification

Equipment shall be provided in accordance with IEC 60079.

The required certification method shall be Intrinsic Safe (IS) with the use of active (galvanic) isolation amplifiers. Zener barriers shall only be used where no alternate active (galvanic) isolation amplifiers exists and agreed with company

Flameproof (EEx d) shall be used where no IS device is available. Purging (EEx p) shall only be used in special cases, e.g. analyser houses and shall be subject to company

approval. Increased safety (EEx e), or intrinsically safe (EEx i) shall be used for junction and terminal boxes.

Solenoid valves shall be Ex d IIC T4-T6 (ATEX-Certified) as per IEC60079

5.4. Electromagnetic Compatibility

Control and instrumentation shall comply with all relevant European Union Electromagnetic Compatibility (EMC) directives and, in particular, compliance with IEC 61000 as follows:

Part 1 - General Introduction

Part 2 - Electrostatic Discharge Require

Part 3 - Immunity to Radiated Electro

Part 4 - Immunity against Fast Ti

Part 5 - Surge Immunity Requirement

Part 6 - Immunity to Conductive Radio Frequency Disturbances

5.5. Safety Integrity Level

All safety instrumentation shall be provided to meet IEC 61508 / 61511 associated requirements including application of multiple measuring and final control elements in combination of voting schemes when required (will be defined during SIL studies).

5.6. Transmitters

Transmitters in general shall be supplied as a head mounted or integral part of the instrument. Where remote transmitters or electronics are required suitable cabling shall be provided to connect the sensor to the transmitter and the parts shall be supplied as a complete unit.

Transmitters shall in general be electronic, 2 wire, 4-20mA, 24VDC loop powered 'Smart' based on Hart protocol. Where 'Smart' instruments are not available for certain applications, standard electronic transmitters shall be used.

Electronic transmitters shall be supplied complete with integral output meter capable of displaying measured variable, e.g. digital LCD indicator.

Where required for operational reasons, remote indicators shall be provided at the specified location.

Where separate transmitters are used to measure the same process variable (e.g. one transmitter for DCS and another for ESD), separate taps shall be installed and transmitters shall have the same range for comparison purposes where deviation alarms shall be configured and displayed in the DCS.

Process connection (transmitter side) shall generally be 1/2" NPTF, electrical cable gland connection M20.

5.7. Flanged Connections

Flanged Connections shall be in accordance with the piping pressure rating with the exception of the following:

- Control Valves and ESD Valves up to 4" inch size the rating shall be ANSI 300 # RF as minimum (To be confirmed)
- Instrument nozzle flange on vessel shall be ANSI 300 # as minimum

Flange surface finish shall be in compliance to ANSI B 46.1

5.8. NACE Requirements

NACE-MR-01-75, latest edition requirements should be applied to wet sour service, where sulphide stress corrosion cracking may occur.

5.9. Instrument Identification Name Plates and Labels.

Name Plates in English and Russian

All field instruments shall be furnished with a stainless steel corrosion resistant nameplate both in English and Russian, permanently fastened with screws and stamped as follows:

- Manufacturer's name, model number, serial number, tag number, supply voltage, range, output, etc. For safety application (ESD) the label shall have red colour.

Service Description Labels in English and Russian

Additional labels for all field instruments to show, both in English and Russian:

- Instrument Tag Number, service description, warning notes as required.
- For safety application (ESD) the label shall have red colour.

5.10. Standardisation

General

C&I equipment for the project shall be standardised. Standardised supply shall include:

- Field instrumentation including Solenoids valves and Limit Switches
- Cables, cabling material and associated junction boxes
- Instrument tubing
- Instrument Fittings

The above C&I standardisation philosophy shall apply to all packaged equipment.

Any exception to the above related to specialised mechanical equipment shall be subject to Company approval.

6. INSTRUMENT UTILITIES

6.1. Electrical Power

Control and instrumentation power supply shall be in accordance with Control Systems Philosophy and Specification 108184-0000-IA-SPC-0001.

6.2. Instrument Air

Instruments air will available (To be confirmed) as follows:

Normal pressure: 7.0 Bar(g)

Min : 3.5 Bar(g)

Max: 9.0 Bar(g)

Temperature °C: -40 to 48

Dew point °C: -50° at 7 Bar(g)

7. INSTRUMENT AND ELECTRICAL INTERFACE

For the instrument and electrical interfaces refer to Control Systems Philosophy and Specification (to be developed by Contractor).

Interposing relay cabinet shall be installed in the electrical substations.

8. INSTALLATION REQUIREMENTS

8.1. General

Contractor shall be responsible for providing a detailed list of instrument installation activities, together with bills of quantities and applicable drawings associated with instrument installation.

Attention shall be given during construction to the protection of equipment against general construction site hazards and in particular the protection of equipment from the weather. Instruments and panels that cannot be installed upon delivery should be housed in a properly constructed air conditioned / heating (as necessary), building(s) protected from dust and damp.

Whenever any protective coated pipe, cable tray, supports steel, etc. requires cutting for field installation, Contractor shall ensure that all cut ends are carefully de-burred, made smooth and finished in accordance with good construction practice.

Contractor shall ensure that all fixings are compatible with the material that they are connecting to/supporting. Where necessary Teflon or equivalent insulating strip shall be fitted to ensure isolation of dissimilar materials, particularly stainless steel and galvanised steels.

8.2. Instrument Location and Support

Installation shall be in accordance with the following guidelines

- Basis of design
- 'API 550' Manual on Refinery Installation of Instrument and Control System.
- 'API 551' Process Measurement Instrumentation
- 'BS 6739' Instrumentation in Process Control Systems: Installation and Design Practice.

All instruments shall be installed in such a way that they are not subject to shock, excessive vibration or temperature extremes, and in accordance with the manufacturer's instructions and good practices.

Electronic field instruments shall be protected in heated boxes with electric thermostat-controlled heaters.

Electronic field instruments exposed to direct sunlight shall be provided with sunshades.

All instruments shall be correctly leveled and aligned before fastening to the supports. Instrument supports shall be of stainless steel material, including U bolt and bracket.

Instruments shall be located with due regard to accessibility and maintenance activities. Vents, drains and filling points shall be sited to minimise the impingement of process fluids on instrument equipment and cables.

Instruments shall be installed to allow for ease of operation, testing, and removal for shop calibration and maintenance purposes.

Instrumentation requiring operator access and post mounted field instruments shall be installed with the centreline between 1200-1500mm above the platform or grade

All instrument equipment shall be accessible for servicing from floor level or from a permanent platform.

9. CABLING REQUIREMENTS

9.1. General

Instrument cabling shall be installed with due respect for safety, reliability, access, maintenance, etc. The drawings, specifications and material requisitions developed for the project will be prepared by Contractor in accordance with the basis of design as outlined below.

9.2. Design Philosophy

All cables shall be as specified in INSTRUMENT CABLE MATERIAL STANDARD (to be provided by Contractor). This specification shall provide details of cable type numbers, cable construction, signal categories, signal groups and service applications.

Cables and multi-cables for above ground routing shall be provided with armour.

Multicore cables shall be sized such that they contain a minimum of 20% spare conductors at the completion of contract design averaged across a given plant.

Junction boxes shall be dedicated to one signal category only, and provide one homerun cable for one JB

Junction box materials shall be 316 stainless steel.

Intrinsically safe signals shall be connected to Eex"i" junction boxes. Those junction boxes shall be provided with terminals and cable glands in light blue colour. Cable glands type shall be M20.

Non-intrinsically safe signals, like solenoid valves, shall be connected to increased safety junction boxes (Eex"e").

Intrinsically safe circuits and non-intrinsically safe circuits shall not be contained within the same cable, marshalled in the same junction box and terminated in the same marshalling cabinet.

Intrinsically safe barrier shall be installed in the marshalling cabinets.

All cabling shall be suitably protected against mechanical damage, chemicals and heat.

9.3. Cable Separation of Instrument and electrical power cables

Main cable routes are to be provided by Contractor on detailed drawings and these shall take into account the separation and segregation requirements of Instrument cables from electrical power cables. These requirements are as indicted below

Instrument cables shall be routed separately from electrical power cables. The physical separation of signal and power cables on parallel runs shall be as below:

Power Wiring Voltage Minimum Separation (mm)

Up to 125V at 10A	250
125V to 250V at 50A	500
250V to 440V	750
6/11KV	1500

Crossovers that bring signal and power cables into close proximity shall be made at right angles. The minimum separation at the point of crossover is 250mm.

The requirements described above may be relaxed at entries into instruments and panels.

Instrument cables carrying more than 10A shall be treated as power cables.

Instrument power cables carrying 230VAC, shall be run on trays separate from instrument signals trays.

I.S. and non-I.S. cables shall not be run together in the same cable trays or cable ducts. In a direct buried trench they must be separated into different bundles and laid as far apart from each other as possible.

9.4. Cable Installation - General

Each cable shall be run in a continuous unbroken length. Cables shall only be terminated in instruments, junction boxes or other approved equipment. No intermediate cable joints are permitted.

All unused entries in junction boxes and other instrument equipment shall be plugged using manufactured, certified as necessary threaded plugs.

When installing cables the radius of all bends shall exceed the minimum bending radius specified by the cable manufacturer.

Installation, terminating, jointing and testing of fibre-optic and co-axial cabling shall be strictly in accordance with the manufacturer's instructions.

Above ground cables shall be fixed using UV stabilised PVC covered stainless steel strapping. PVC cable ties shall not be used. Cables shall be fixed at approximately 500mm intervals in vertical runs and 1000mm intervals on horizontal runs.

Cables connected to instruments shall be installed with a sufficient length of cable to provide sufficient slack for re-making the cable connection if the instrument is removed.

9.5. Underground Cable Installation

Above ground is the preferred method for cable routing. Where unavoidable or crossing roads then relatively short straight routes can be run underground in cable ducts.

9.6. Above Ground Cable Installation

Cabling between Rack Rooms, Junction Boxes and individual instruments shall be routed above ground wherever possible and shall be supported by a suitable cable support system. Cable Ladder/Cable Tray shall be used for the main routes and Cable Tray for local spurs of up to 3 single pair or triad cables. The main cable routes and some of the smaller routes shall be shown on the detail drawings.

The following general requirements are applicable to all above ground cable installations:

Mechanical protection of above ground multi pair cable shall be achieved by using steel wire armour cables routed on galvanised Cable Ladder/Cable Tray.

Cable trays shall be grounded at the beginning and end of each continuous section.

Cables may be stacked 2 high (max) on Cable Ladder/Cable Tray installations. Spare capacity shall be provided on all installations (1 additional cable or 25% of total number, whichever is greater).

Cable separation shall be maintained in above ground cable installations. Instrument power cables shall not be run on the same tray as signal cables.

For dual redundant fibre optics cabling as required for the control, monitoring and safety systems for the plants, connecting the various buildings. The cables shall be run on separate routes with the primary cable run over head and the secondary cable route run under ground in ducting. This to ensure maximum separating and safety of the communication link in the event of damage to a fibre optic in any route.

Cables crossing canals, ditches, pipe trenches etc., shall be protected by a bridge carrying cable tray of appropriate width.

9.7. Cable Installation Inside Buildings

Cables within control building/rack rooms etc. will normally be routed under a raised computer floor. Where cables are routed above the floor (fire system associated cables) they shall be in conduits. The main routes shall be shown on the detail drawings.

Incoming field cables shall be routed directly on the floor underneath the raised computer flooring. Internally routed cables under the access floor shall be run in heavy duty cable tray.

The cable tray shall be sized with spare space (30%) for additional cables.

The cable tray system and supports shall be constructed of hot-dipped galvanised steel.

The cable tray system shall be supported off the floor.

Apart from field cables which are laid directly on the floor there shall be two additional cable tray routes, one containing AC power and grounding cables, the other containing system cables, communications cables etc...

System cables shall be of the shortest standard length that allows the correct location and termination of the equipment. Spare length shall be neatly coiled in the tray below the system cabinets.

Fragile cables (fibre optic pigtails, etc.) shall be run in flexible conduit.

Conduit shall be free from burrs and be neatly run and adequately supported. PVC conduit shall avoid equipment operating at elevated temperatures.

Cables entering a building shall do via O.D 150mm PVC pipe bank from the bottom side of the buildings, below the raised floor. Cable trays local to transit frames shall be sloped downwards away from the building to ensure that any liquid is directed away from the cable transit.

All the multi-cables entering in the buildings should be sealed at the entry, after their installation, to prevent entering of water, moisture, flammable gases.

9.8. Fire Protection

All Instrument cable shall be flame retardant according to IEC 60332-3.

Instrument signal wiring that is identified as being part of a fire and gas protection system or part of deluge/extinguishing system that must continue to operate normally during a fire (fire critical) shall be made fire resistant. The preferred wiring system for these circumstances is to use fire resistant cables according to IEC 60331. Junction boxes as part of these systems shall be where possible mounted outside the fire risk area. Where this is unavoidable then the JB shall be fire-protected.

Testing methods for fire retardant cables should be as per local standards.

9.9. Cable Glands

All cables in the field shall be equipped with glands.

All cables terminated at the equipment/rack room in floor mounted cabinets do not require glands but shall be securely supported at the base of the cabinet using a cable clamping bar; cables terminating at wall mounted cabinets shall be equipped with glands.

Cables shall be terminated in a safe neat manner using approved nickel plated brass compression glands. SWA cables shall have all strands of the wire armour securely clamped in a permanent manner.

All gland installations shall maintain the IP 65 rating. Sealing washers are to be provided at entry to each field instrument installation.

The gland thread shall be metric thread match that of the associated equipment wherever possible. The use of thread adaptors shall be minimised.

Cable glands in the field shall be protected by fitting PCP shrouds.

All glands shall be fitted in the bottom of equipment located in the field. Any side cable entry if necessary shall be agreed with Company. Top gland plates shall not be used. Any unused cable entries shall be sealed using a proprietary plug, certified as necessary to suit the equipment and location.

9.10. Wiring Terminations

Wiring terminations shall be in accordance with the following general requirements:

All terminals shall be screw clamp type where the screw is not in direct contact with the wire. Terminals may incorporate test sockets if necessary. Power supply terminals and all terminals with voltages over 50V shall be fitted with protective covers and warning labels. Power supply terminals shall be fitted with partitions between adjacent terminals.

Solid conductors if used shall not be fitted with crimped connections. Stranded conductors shall be crimped using a crimp which also grips the insulation of the conductor. The termination end of the crimp shall be suitable for the terminal to which it is connected. All crimping shall be carried out using the crimp manufacturer's approved tool. Only one conductor shall be terminated in each terminal.

Sufficient slack wire shall be left looped at terminals to allow for re-making of terminations.

All spare cores in multicore cables shall be terminated. Sufficient terminals shall be fitted in all equipment to facilitate this.

Each terminal shall be clearly and permanently identified. When there is more than one terminal strip in an enclosure, then each strip shall be clearly and uniquely identified. For example in a marshalling cabinet the strip name will be the same as the cable number.

10. EARTHING SYSTEM

For equipment earthing and bounding of field instruments, junction boxes, local panel, cabinet inside building, three different type of earth shall be installed in building (satellite room or CCR).

The three earthing system shall be provided fully segregated. Connection of IE bars and ISE bars to the main PE system shall be realised by the electrical department.

10.1. Instrument Earthing (IE)

Instrument Earthing system shall be used for voltage reference of instruments, electronic system, cable screens, which are not intrinsically safe. These equipments shall be connecting at only one point inside buildings located at the bottom of each marshalling cabinet.

On field instrument side, cable screen shall be insulated and floating. At least the measured resistance of IE shall not exceed 1 ohm.

10.2. Instrument Protection Earthing (IPE)

Instrument Protection Earthing system (mechanical earth) shall be used for protection of personnel. All mechanical portions like supports, cable trays, frame junction boxes, local panel, etc..shall be connected to the IPE. At least the measured resistance of IPE shall not exceed 5 ohm.

10.3. Intrinsically Safe Earthing (ISE)

As a general rule all the instruments installed in hazardous area shall be intrinsically safe and connected to the ISE system at only one point inside buildings located at the bottom of each marshalling cabinet. At least the measured resistance of ISE shall not exceed 1 ohm.

11. INSTRUMENT PROCESS PIPING AND TUBING

11.1. General

Tubing and fittings shall be in general 1/2"outside diameter, duplex 2507 (or 2205) stainless steel, seamless and annealed, a higher grade of materials shall be used where process conditions dictate.

Tubing compression fittings shall have NPT threads. Stainless steel tubing shall have stainless steel fittings.

Steel piping shall be jointed with approved fittings as specified on the drawings. Where practicable, pulled bends shall be used in preference to fittings.

Due regard shall be given to cold/hot service installation that is subject to expansion/contraction, to ensure that sufficient flexibility is allowed.

11.2. Instrument Air Piping

The location of the main instrument air headers and the take off points shall be shown on the installation drawings.

Instrument Air Manifolds shall be used in areas where there is a high density of users.

Where user density is low a spider system of sub-headers terminating within 1 metre of the individual consumer shall be used.

Design shall allow for a minimum of 20% installed spare user capacity for future use.

Air supply headers shall be sized as follows:

1-5 users - 1/2 " NB

6-10 users-3/4 " NB

11-25 users - 1" NB

All branch take-offs shall be from the top of the horizontal main header and main header shall slope to drain.

All take-offs from the main air header shall have an isolation valve.

Drain legs are supplied at each low point of the

11.3. Instrument Air Supply & Signal Tubing

Instrument air supply from main air header take-off valve to air manifold, or if spider to within 1 metre of the consumer, shall be run in galvanised carbon steel pipe, the final run from the air manifold or end of the galvanised carbon steel pipe to the filter/regulator and instrument shall be in 3/8" or 1/2" duplex 2507 stainless steel tubing with duplex 2507 stainless steel compression fittings.

11.4. Piping and Tubing Supports

One or two single pipes or tubes maybe supported by fastening to local structural steel or pipe stands, provided the local ambient temperature is not excessive and tubing does not interfere with other equipment. Tubing shall not be fastened directly to process lines or other process equipment. Vibrating structures and equipment shall be avoided.

Care shall be taken to avoid stainless steel tubing coming into direct contact with galvanised supports.

12. SPECIAL MEASUREMENT APPLICATION

For special instrumentation, these instruments that are required by Process Licensor(s) and detailed on their Recommendations are considered mandatory and shall be engineered, designed, supplied and installed in accordance with the Process Licensor(s) Recommendations.

These instruments may be as follows, but not limited to:

- Alky analysers
- Process analysers for composition
- Measuring of contact and activator
- In-line equipment of suspension fluids (flow, valves, etc)
- Valves for wax application
- Flow measurement for wax application.
- Temperature measurement of melt in the extruder
- High pressure values

- Etc.

13. FLOW INSTRUMENTATION

13.1. Orifice Plates

Orifice plates shall be the primary measuring device for differential pressure transmitters used for liquid, vapour and gas services.

Manufacturing of orifice plates shall be generally in accordance with ISO 5167 standard.

Concentric, sharp edged orifice plates shall be used in most applications. The use of eccentric orifices shall be limited to gases containing liquid, liquids containing solid particles, or liquids containing gas. The use of segmental orifices shall be limited to vapours containing liquids, liquids containing solids, or liquids containing gas.

Orifice plate material shall generally be stainless steel AISI 316. Other higher grade materials shall be selected to suit the process fluid and the flowing process condition.

A precision pipe assembly with orifice plate shall be used for line sizes with nominal diameter less than 2".

Orifice tag number, material, direction of flow, nominal diameter and bore diameter shall be permanently stamped on the orifice plate handle.

The calculation of orifice plates shall be in accordance with ISO-5167 standard.

Range of pressure drop of flow meters shall generally be 0 - 250 mbar.

The d/D ratio shall normally be between 0.25 and 0.75.

Orifices shall be sized so that the normal flow rate falls at approximately 70% of maximum chart and minimum flow not less than 30% of minimum chart.

Orifice plates shall be used only for line sizes 2" up to 24" (if the line size is less than 2" an integral meter run shall be used).

Drain hole for vapour service and vent hole for liquid service shall be provided for the orifices in horizontal runs, as specified on the individual data sheets.

The orifice flange shall be weld neck type as per ANSI B 16.36 complete with pressure taps.

The connections (1/2" or 3/4") for the pressure taps on the orifice flanges shall be in accordance with the piping specification. Thread tapping should be minimized.

Pipe tapping where specified shall be applied and the orifice plates shall be installed into the normal flanges in accordance with pipe class specification.

13.2. Venturi Tubes

Manufacturing of venturi elements shall be generally in accordance with ISO 5167 standard.

Venturi or low loss flow tubes shall be used where good pressure recovery is essential. Applications are on liquids, gases and steam.

Flow range is limited only by minimum and maximum beta ratios.

Materials for venturi tubes shall be in accordance with piping specification.

Minimum of four pressure tapings on each annular chamber for Venturi tube shall be provided for the entrance section and the throat section.

Pressure tapping for venturi tubes to be (1/2" or 3/4") Thread tapping should be minimized. 3" Flanged pressure taps with diaphragm seal shall be used for large Venturi's for services on the reactor recycle gas compressor.

13.3. Restriction Orifice Plates

Restriction orifice plates shall be constructed with a thickness specified at full rating. The thickness shall be suitable for the process fluid and the process flowing conditions.

Restriction orifice plate material shall generally be stainless steel AISI 316. Higher grade materials shall be selected to suit the process fluid and the process flowing conditions.

Drain/weep holes shall not be provided on the restriction orifice plates.

Orifice bore shall not be bevelled for the restriction orifice plates.

Each restriction orifice plate shall be provided with a corrosion resistant tab that projects beyond the flange. The tab shall be stamped with the material, bore, thickness, I.D. of the pipe and identifying tag number.

13.4. Positive Displacement Flow Meters

Positive Displacement Meters (PD Meters) shall be used for liquid services for measuring volumetric flow where integrated flowing quantity of high accuracy is required.

The meter body configuration shall be constructed to minimise pressure drop through the PD Meter.

Straight through strainers with suitable mesh size, material, and rating shall be installed preceding each meter.

The accuracy of measuring device shall be at least 0.1% of full flow meter range. Linearity of 0.1% over a 5 to 1 range, repeatability of 0.02% over rated flow range.

Rotor bearing shall be of low friction stainless steel and should not require lubricating. PD Meters shall not be used with non-lubricating liquids such as propane and butane.

PD Meters and their associated strainers, air-eliminators, and flow regulators shall have ANSI flange process connections to ANSI B16.5 in line with the pipe specification.

Flow direction shall be clearly and permanently stamped on the body of the meter.

13.5. Turbine Flow Meters

Turbine meters shall be used in relatively clean liquids, gases and vapour services. The rangeability of these meters is typically 10:1. Turbine meters shall be selected to suit the process fluid properties such as viscosity, density, etc.

The accuracy of the turbine meter shall be 0.1% of flow rate and the repeatability shall be 1%. Linearity shall be 0.1% over a 5 to 1 range and repeatability of 0.02% over rated flow range.

Minimum of two hermetically sealed or potted, pickups shall be provided for each flow meter.

Turbine Meters shall be designed for at least 130% of the flow range specified.

Each Turbine Meter shall have a nameplate indicating the average pulse rate over the meter range specified.

Strainers with the mesh size, material, and rating shall be installed preceding each meter.

Turbine Meters and their associated strainers, air-eliminators, and flow regulators shall have ANSI flange process connections to ANSI B16.5 in line with the pipe specification.

Flow direction shall be clearly and permanently stamped on the body of the meter.

Meter accessories shall include but not limited to the following:

- Signal converter
- Flow straightener
- Read out electronics

13.6. Variable Area Flow Meters

Variable Area Flow Meters (Rotameters) shall only be used on minor services such as analyzer sample conditioning and purging and shall not be used on process duties. Glass tubes may only be used for purging installation at low pressure and not dangerous fluid (e.g. air, nitrogen, etc).

All variable area meters shall preferably be metallic tube, flanged and provided scale graduated in engineering unit.

13.7. Magnetic Flow Meters

Magnetic flow meters shall be used for volumetric flow of conductive liquids, including slurries and corrosive or abrasive materials. The minimum fluid conductivity required is between 1 and 5 microS/cm.

Bipolar pulse type magnetic flow meters with low power consumption shall be used. Power supply connections shall be made generally to the flow converter only. A power-isolating switch shall be provided to facilitate local power isolation.

Where process fluids could be abrasive, counter flanges shall be provided with detector body to prevent damage to the internal lining. Construction of coil and internal lining shall be suitable for maximum rated temperature of the fluid.

The detector body shall have flanged connections to ANSI B16.5.

The accuracy of the meter shall be 1% of full scale and the repeatability shall be 0.25%.

The meters installation shall assure that the meter body remains full of liquid.

Flow direction shall be clearly and permanently stamped on the meter body.

13.8. Vortex Flow Meters

Vortex Flow Meters shall be utilized, where the required flow meter ranges are very wide and the orifice-type flow meters cannot be applied.

Vortex meters are suited for clean liquids, gases, and steam services and shall not be used on services that are dirty, abrasive, viscous, or mixed flow (gas with liquid droplets, liquid with vapour bubbles).

Vortex meters shall not be installed in piping if they are subject to vibration.

Vortex meters shall have ANSI flange process connections to ANSI B16.5 in line with the pipe specification.

Flow direction shall be clearly and permanently stamped on the meter body.

13.9. Ultrasonic Flow Meters

Ultrasonic flow meters shall be the Transit Time / time of flight type. Single or multiple beam designs shall be considered depending on the application. The greater the number of beams used, the less the sensitivity to velocity profile effects.

Laminar or turbulent flows can be measured using the double path instrument.

Ultrasonic flow meters shall be used for fluid measurement where no loss of pressure can be tolerated.

Ultrasonic flow meters shall be considered for clean liquids without solids or bubbles, and for gas services. Meters selected shall be suitable for the flowing fluid velocities limits.

Flow meters with insertion probes shall have retraction mechanisms to allow on stream probe maintenance.

Clamp on type Ultrasonic flow meters shall be only used where specified

Flow meters shall have ANSI flange process connections to ANSI B16.5 in line with the pipe specification.

The accuracy of the meter shall be 0.5% of full scale or better.

Flow direction shall be clearly and permanently stamped on the meter body.

13.10. Pitot Tube

Pitot tubes (annubars) may be utilised in large pipe diameters associated with secondary services (non process related) and where accuracy is not a critical factor. The use of any Pitot tube shall be subject to Company agreement.

Pitot tube length and active length shall be designed to suit the pipe diameter, and in cases where deflection or oscillation may occur, it shall be supported or secured at the opposite side.

Pitot tubes shall have minimum 3" flanged connections. Flange rating and material shall be as per the pipe specification.

Tube material shall be AISI 316 stainless steel as a minimum.

The head shall have 316 stainless steel connection block with 1/2"NPT (F) ports.

Instrument connections shall be supplied complete with 1/2"NPT stainless steel valve and nipple.

13.11. Coriolis Mass Flow Meters

Where a high accuracy is required, coriolis meters shall be used to measure liquids, slurries, compressed gases, raw materials, utilities and liquefied gases in line sizes up to 6". Gas/liquid mixtures shall be avoided.

The accuracy of the meter shall be 0.5% of full scale or better, and the repeatability shall be 0.25%.

The rangeability of coriolis meters shall be typically 20:1.

Coriolis meters shall be capable of measuring mass flow rate, volumetric flow rate, fluid density and temperature from the same sensor. Measurement shall be insensitive to changes in temperature, pressure, flow profile and fluid viscosity. The transmitter shall be smart type with analog output 4-20 mA configurable for flow rate, density or temperature.

General installation shall ensure that the sensor is full of process fluid. When measuring liquids, the sensor must not be installed at a high point, as any gas may accumulate in the sensor. Similarly, when measuring gases, the sensor must not be installed at a low point in the system, as any condensation may accumulate in the sensor. If the meter is installed in a vertical line, the fluid flow shall be upwards through the sensor. Upstream and downstream straight run requirements and flow conditioning are not necessary, due to independent flow characteristics.

The body and sensor materials shall be AISI 316 stainless steel as a minimum. Other alloys shall be used as per process fluid requirements.

Coriolis meters shall have ANSI flange process connections to ANSI B16.5 in line with the pipe specification.

Flow direction shall be clearly and permanently stamped on the meter body.

13.12. Thermal Mass Flow Meters

Thermal mass flow meters shall be used to measure gaseous fluids that have constant composition. The main application is in measurements of gas flows in quality measuring systems and purge flow measurement to processes. Gas velocity shall not exceed 60 meters/second.

Normal application is in pipe sizes or ducts up to 3".

Thermal mass flow element process connection shall be flanged to ANSI B16.5. Flange rating and material shall be as per the pipe specification.

The flow element design shall allow for field retractable packing gland assembly.

All wetted surfaces shall be 316 stainless steel, as a minimum.

The accuracy of the meter shall be 0.5% of full scale. .

Flow direction shall be clearly and permanently stamped on the meter body.

13.13. Venturi Nozzles

Venturi nozzles shall be generally in accordance with ISO 5167 standard.

High beta nozzles are recommended for diameter ratios between 0.45 and 0.80.

Low beta nozzles are recommended for diameter ratios between 0.20 and 0.50.

The nozzle shall be supplied complete with a holding ring for mounting between flanges.

Materials for Venturi nozzles shall be in accordance with the project piping specification.

13.14. Flow Switches

Use of Flow switches is generally not recommended and shall only be used where no other alternatives exist and shall be subject to Company agreement.

14. TEMPERATURE INSTRUMENTATION

14.1. Temperature Sensing

Temperature transmitters shall be used for all temperature measurements except bearing and winding temperatures. All temperature transmitters shall smart digital with 'Hart' protocol.

Integral head mounted temperature transmitters are preferred; however remote mounted transmitters shall be provided the following:

- Skin measurements types
- Multi element types
- High vibration areas

Where suitable, temperature detection shall be via PT100 Resistance Temperature Detectors (RTD). Thermocouples may be used where RTD's are not suitable, such as high temperature range, the presence of high frequency vibration etc.

For local indications Bi-metallic every angle dial thermometers shall be used. Where Bi-metallic types are not suitable, gas or liquid capillary instruments may be used.

Mercury filled systems shall not be installed.

14.2. Thermowells

Thermowells shall be provided for all process temperature measurements with the exception of 'skin type'. Minimum line size for thermowell installation shall be 4". Pipe lines that are smaller than 4" shall be enlarged to accommodate the thermowells installation.

Generally, 2" flanged thermowells shall be provided on vessel and 1.5" flanged thermowells on pipe in accordance with pipe lines ratings

Screwed type wells may be considered for minor services on small pipe lines where it is not possible or practical to install a flanged thermowell.

However this shall be agreed with Company on a case by case basis. These shall have external thread size of 1" NPT male with 1/2"NPT female internal thread and shall withstand a minimum of 20bar at 200 C°.

Thermowell material of construction shall be 316SS as a minimum.

Natural frequency and wake frequency calculations shall be performed as per ASME PTC 19.3.

Well material, nominal well length, rating and tag number shall be stamped on the side of the flange or on the hex flat for screwed type wells.

14.3. Resistance Temperature Detectors

Resistance temperature detections (RTD) and transmitting or receiving instruments shall be 3-wire RTD's shall have dual elements, platinum wire type, 100 ohm @ 0 degC Class "A"

The use of resistance thermometers is normally limited to the range -200 to 450°C.

Elements shall be connected to the terminal head by means of loading springs to assure a permanent contact between sheath and thermowell.

14.4. Thermocouples

Thermocouples shall be mineral insulated, sheathed to 6mm diameter in accordance with IEC 60584 and with hot junction insulated from the sheath.

Thermocouple head terminals shall be marked with positive and negative symbols.

For temperatures range above 450 °C or where mechanical vibrations are suspected to be possible. Thermocouples shall be used and manufactured from nickel/chromium Vs nickel aluminium (Type K) wires with reference characteristics in accordance with IEC-60584.

For temperatures above 1400 °C, or for H2 service above 750 °C, noble metal thermocouples shall be used.

14.5. Burn-out Protection

RTD's and T/C circuits operating high temperature alarms shall have upscale element burnout. Those on low temperature alarms shall fail to down scale position.

RTD's and T/C on SIS duty shall fail to anti-shutdown position

All thermocouple actuated control systems shall contain 'burn-out' protection. The design shall enable this feature to be switched in or out.

15. LEVEL INSTRUMENTATION

15.1. General

Level connections shall be provided directly on the side or top of the equipment, i.e. bottom connections shall not be used.

Bridles shall not be used for flammable or explosive services.

Storage tanks for flammable or explosive services shall be provided with two level instruments that operate on different principals.

15.2. Local Indication

The following local level instrument types shall be used:

- Gauge glasses for vessels and small tanks. Gauge glasses shall be provided where indicated on the Process Licensor Design Recommendations.
- Float type instruments for large tanks, where local indication only is necessary or differential pressure instruments where float type instruments are not suitable (e.g. viscous fluids).
- Magnetic metal type indicator shall be used on hazardous process fluids.

Local level indication where specified shall span the full range of the continuous measurement instruments including any alarm and trip points.

15.3. Continuous Measurement

All continuous level transmitters where applicable shall be smart type with 'Hart' protocol.

Differential pressure instruments shall be used as the preferred choice for all level applications except where indicated otherwise.

Displacer type instruments may be used for interface measurement where the range does not exceed 1219mm. The standard ranges of 356, 813 and 1219 mm shall be used.

Other types of measurement, e.g. R.F. admittance, capacitance, and ultrasonic may be used subject to agreement with the Company where differential pressure instruments or displacers are unsuitable.

15.4. Local Gauge

Magnetic type level gauges are the preferred for the local level measurements for vessels, reactors, tanks where required.

Gauge glasses if considered these shall be 'transparent' or 'reflex' type with chambers and covers made of carbon steel and heat resistant borosilicate glass. Alloy or 316 stainless steel construction shall be used for all wetted parts where process requires it.

Transparent gauges and reflex gauges shall be fitted with toughened glass.

Tubular glass type gauges shall not be used, except where specified by a Process Licensor.

Reflex gauge glasses if considered shall be used on clean services.

Reflex gauges may be used for liquid and vapour interface detection except in the following circumstances, where transparent gauges shall be used: -

- Interface between immiscible liquids.
- Viscous, caustic or acidic service

- Colour or turbidity observation.

All gauges shall be supplied with a shut off valve on the top and bottom mountings and a full bore drain valve. A vent valve shall be provided on toxic services or on corrosive liquid and liquid interface duties to allow for piping for safe fluid disposal. A vent plug shall be provided where a vent valve is not fitted.

All gauge glasses shall be provided with ball check valves to prevent fluid discharge in the event of glass breakage. Level gauge length center to center of gauge end-connections shall not be more than 1570 mm. To cover larger ranges two or more gauges shall be used with a minimum 50 mm of overlap on the visible glass.

Each gauge shall be stamped with the maximum working pressure and temperature that is equal to or higher than the vessel design pressure and temperature ratings.

The materials to be used for valves, the fittings, and the components of gauge glasses in contact with process fluids shall be suitable for the process conditions. Vent and drain connections shall be 3/4" NPT female or flanged as per the vessel trim. Seal welding shall not be provided on these connections.

Vessel connections shall normally be 2" flanged.

15.5. DP Level Instrument

The level transmitters shall be provided with full elevation and suppression availability to have the increasing/decreasing output signals according to the increasing on the liquid levels or the interface levels.

The size of the flange for diaphragm seal type applications shall be 3" as a minimum.

All internals in contact with the process shall be 316 SS as a minimum. High grade materials shall be provided where process conditions dictates.

Accuracy of all measuring devices shall be at least 0.5% of full span.

15.6. Displacer Type Level Instrument

Displacer type instruments, where specified for use, shall normally be carbon steel bodied with stainless steel displacer and Inconel torque tube unless higher-grade material is required.

Displacement level instruments shall be provided with a minimum of 2" flanged connections in accordance with ANSI B16.5.

Displacer transmitters shall be electronic and shall be installed in the instrument cases associated with the displacer cages. Maximum error shall not be greater than 0.5% of span. Hysteresis shall not be greater than 0.2% of span. A means of correcting for specific gravity shall be provided.

Internal displacers may be used on vessels where an external arrangement is not possible (e.g. sumps). Where the displacer is subjected to turbulence, the effect shall be minimised by shielding, guidance or other mean

15.7. Level Switches

Level switches shall not be used except where no other alternative measurement exists and shall be subject to Company agreement.

15.8. Nuclear Level Measurement

Nucleonic instruments will consist of radioactive source and detector (s) mounted on opposite ends of vessels or pipes.

Nuclear level measurement shall be used when no other technology is available particularly in reactors and fluidized filters for the polymer units and where required on other unit. Where a large span of level is required to be measure several detectors can be used in combination with appropriate DCS algorithm to cover the required level span.

Nucleonic instrument are also used to measure density of the slurry.

Radio active sources shall be preferably Caesium 137, Cobalt 60 or Americium 241, depending on the respective application.

The sources shall be encapsulated in lead-filled steel safety holders provided with on/off shutter lockable in the off position.

The installation of equipment shall be in accordance with the Republic of Kazakhstan laws and regulation.

15.9. Tank Level Measurement

The following tank level instrumentation shall be provided for the complexes that are suitable for continuous service under normal conditions.

Installation of the field instruments and equipment shall allow for easy maintenance without the removal of the equipment from the tank.

Radar Gauging

The preferred tank gauges shall be Radar devices.

The radar unit shall comply with the following requirements as a minimum.

Accuracy shall be to 2 mm or better

Power supply shall be 24 VDC.

The unit shall be self-supporting type and be suitable for flange mounting on top of the tank.

The tank gauges shall be provided with local indicator mounted at grade.

The Radar gauges shall be connected via serial links to form the tank gauging system as outlined in the Control Systems Philosophy and Specification 108184-0000-IA-SPC-0001.

Where process vessels or tanks utilizing radar level measurement not forming part of the tank gauging system but employed to measure the level for process purposes. The output signals from of the level gauges shall be 4-20 mA connected directly to the DCS to display the liquid levels in the tanks.

The above shall also apply to the servo gauges below.

Servo Gauges

The servo gauges shall be servo/float type, spring balancing mechanism and shall incorporate a level sensor consisting of a float following the liquid surface in the tank to get 2 mm accuracy as a minimum or better. The repeatability of the instrument shall be 0.5 mm.

The gauge shall be powered from a 230 VAC UPS supply.

The tank gauges shall be provided with a local indicator mounted at grade.

The servo gauges shall be connected via serial links to form the tank gauging system as outlined in the Control Systems Philosophy and Specification 108184-0000-IA-SPC-0001.

Temperature Gauging

A multi-temperature sensor device shall be used for the average product temperature and the product temperature profile in a tank where specified. Each bulb shall have a minimum of 10 resistant elements covering different heights for accurate measurement. The accuracy shall be 0.5 °C as a minimum. A Pt-100 reference element shall be used.

Sheathing and fitting materials shall be suitable for the product and the operating conditions, minimum AISI 316 SS shall be used.

The temperature measurement shall be used in conjunction with the level measurement in the tanks to provide a true tank inventory.

The resistive element terminals shall be brought out into a connection box fitted on top of the element.

The temperature gauging shall also form part of the tank gauging system outlined in the above mentioned philosophy

15.10. Bubblers

These may be used for local or remote indication with Company agreement on open tanks or sumps containing corrosive, slurry, or viscous process liquids.

Expected accuracy for these installations is 1% to 2% of full scale.

The dip pipe shall be at least 2" in diameter in order to minimise the pressure drop as the air flows through it. Two separate connections shall be made to the dip pipe, one for the regulated air supply, the other for the level indicating pressure gauge. The pipe tip shall be kept about 3" from the bottom of the tank to keep it from clogging by sediments.

The purge supply pressure shall be at least 1 Bar (g) higher than the highest hydrostatic pressure to be gauged. The purge flow rate shall be kept small, approximately so that there is no significant pressure drop in the dip pipe.

16. PRESSURE INSTRUMENTATION

16.1. General

All pressure devices shall be capable of withstanding up to 1.3 times the design pressure and test pressure of the equipment on which it will be mounted.

Ranges for pressure instruments shall be approximately 1.5 times the normal operating pressure. However, the selected range must always withstand, without zero or calibration change, the safety relief pressure on the system.

Pressure instruments on corrosive service shall be protected by a diaphragm seal. Diaphragm seals shall be close coupled, flange bolts seal locked, the entire assembly fully supported and protected.

Pressure instruments installed in pulsating services and suction/discharge of reciprocating pumps or compressors shall be provided with pulsation dampeners.

All pressure instruments shall be provided with manifolds (isolation and vent). The material shall be AISI 316 stainless steel as a minimum.

16.2. Local Indication

General

Gauges for the measurement of differential pressure shall be of the bellows or diaphragm type.

Draught gauges may employ a quadrant edgewise indicator of suitable size.

Local indicating pressure gauges shall be suitable for the environmental conditions and shall be readable from grade or platform. Gauges shall have stainless steel movements and in all cases shall be constructed in shatterproof stainless steel

All gauges shall be equipped with blowout discs and be so installed such that mounting arrangements will not restrict the function of these blowout discs.

All gauges shall have shatterproof glass.

Dials shall have a diameter of 100mm. The scale shall be white with black lettering.

Bourdon tubes shall normally be 316 stainless steel. For gauges on corrosive services (i.e. acid, sea water) consideration shall be given to diaphragm seals, Monel construction.

Pressure gauges for severely pulsating services shall have a helical, gearless type movement with micro-range adjustment and dampener. Gauge accuracy to be within 1/2 of 1% of the scale range.

Cases for gauges shall be solid front stainless steel. Cases shall be screwed ring type. Cases shall be vapour tight and weatherproof. The glass shall be shatterproof.

Surface mounted gauges shall have rubber blow out disc located in the lower side of case. Direct and flush mounted gauges shall have rubber blow out disc located in the back of the case. Steam service gauges shall also have a rubber blow out disc in the back of the case.

Diaphragm gauges and diaphragm sealing chambers with features to protect them from chemical attack shall be clearly labeled with the name of the fill fluid.

Diaphragm Seal Gauges

The bottom section shall be removable for cleaning. The entire system above the diaphragm including the element shall be evacuated and entirely filled with an inert liquid. Gauges with diaphragm seals shall have capillary bleeders.

Capillary tube material shall be stainless steel.

16.3. Pressure Transmitters

All pressure and differential pressure transmitters used for control and monitoring and safety application for the complexes shall be electronic, 'Smart' digital with 'Hart' protocol and shall be of the capacitance, strain gauge or resonant wire type.

Generally all parts in contact with the process fluid shall be AISI 316 stainless steel as a minimum. Higher-grade material shall be utilized as dictated by the process medium or conditions. For transmitters on corrosive process applications, diaphragm seals/Monel wetted parts will be used.

For electronic transmitters, circuits shall ensure system continuity in case of local indicator failure or removal.

16.4. Pressure Switches

Pressure switches shall not be used and may be used where no other alternatives exist with Company agreement.

16.5. Diaphragm Seal Systems

Diaphragm seals shall be direct or remote type. All capillary tubing for remote type shall be corrosion resistant and sheathed or armoured.

For general remote type applications, process connection sizes 2" to 4", the pancake type to be used and for smaller process connection sizes 1/2" to 1" flanged, with flange rating in accordance with the pipe specification.

For general direct type applications, process connection sizes 2" to 4", the flanged type shall be used. Rating shall be as per the pipe specification,

Diaphragm and wetted parts material shall be suitable for the process and shall be AISI 316 SS as a minimum.

Lower housing material shall be the same as the wetted parts, minimum AISI 316 SS.

Seal fluid shall be suitable for the operating conditions as specified.

17. LOCAL PNEUMATIC INSTRUMENTATION AND LOCAL CONTROLLER

17.1. General

It is not expected that pneumatic instrumentation or direct field mounted controllers will be employed on the complexes.

However if a particular process requires implementing such instrumentation, these shall be reviewed on case by case basis and shall be subject to Company approval.

18. INSTRUMENT CONNECTIONS

18.1. General

The connection sizes to vessels and/or pipes are the size of connection to be provided for the first block valve and the instrument. If a minimum size connection to the vessel/pipe is required it shall be swaged down without a block valve before the instrument flange.

All connections listed below are flanged unless otherwise stated. Instrument nozzle flange on vessel shall be ANSI 300 # as minimum

It shall be noted by Contactor that the instruments connections, sizes and ratings stated in this document are the minimum and general indication of the requirements for the project.

Contractor shall be fully develop these requirements with Company agreement taking into account the project piping specification, process equipments, process fluids condition, Process Licensor special requirement, etc.

18.2. Connection Size

Interface size shall be review with the Client representative to be consistent during the entire Project.

Instrument Type	Connection To Equipment	
	Vessels/Tanks (Min ANSI 300#)	Pipe*
Themowells	2"	1.5"
Pressure Gauges	1"	1"*(3/4")
Pressure Transmitters	1"	1"*(3/4")
D/P cell (Pressure)	1"	1"*(3/4")
Level Displacers	2"	-
Level Displacers (top)	4"	-
D/P cell Flanged (Level)	2"-4"	-
Level Gauge	2"	-
Level (Radar & Servo)	4-6"	-
Diaphragm Seals	2"-4"	2"-4"

*Piping connection shall be in accordance with piping class specification.

19. MODULATING VALVES

19.1. General

Control Valves used for modulating function shall generally be globe or eccentric rotary plug valves with pneumatic actuators. The minimum size for control valves used shall be 1" or pipe size as per the pipe specification. If a smaller Cv is required, reduced trim shall be used.

Valve bodies with flanged ends shall be in accordance with section 5.7

For applications in sour gas services, material of body, trim and all other wetted metal parts in contact with the sour gas shall be in compliance to section 5.8

Control Valves shall be generally globe type for small size and globe or segmental rotary type for larger sizes. Other types such as characterized ball valves shall be used for high capacity and tight shut-off service. Butterfly shall be used for high capacity and low pressure drop.

Other types such as angle, split body, "Y" should be considered when process may be erosive, viscous or carrying suspended solid.

Globe Valves

The preferred style of control valves is flanged single seat globe type 1" to 4" with the body being of single cast construction.

Double-seated 6" and larger globe valves shall have top and bottom guided construction.

Ball Valves

Ball valves shall be considered for on/off duty or for large sizes on throttling service.

Ball valves shall be considered as throttling valve for hydrocarbon services with coking tendencies, erosive services, or suspended solids where settlement in globe valve body may occur. The use of reduced ball trim is allowed.

Butterfly Valves

These shall not be used for shut off purposes.

60 degree opening butterfly valves shall be considered when the large size valves are required with a low pressure drop or corrosive services where body lining of standard globe valves becomes economically unfeasible.

Angle Body Valves

These may be applied to:

- High noise applications, where a globe valve is not suitable
- Liquid flows where cavitations may occur in the valve
- Hydrocarbon services with tendency towards coking
- Erosive services

Angle valves shall have full venturi throat.

Self Acting Regulators

Self-acting regulators or pressure regulators shall have the same general requirements as control valves.

Reducing regulators, backpressure regulators and differential pressure regulators shall be as required by the process conditions and indicated on the Licensor design Recommendations.

The diaphragm for these shall be designed to meet the design pressure requirements of the process conditions.

End Connection

The Flange ANSI rating class shall be in accordance with section 5.7

Lugged wafer type valves maybe used when indicated in piping specification.

19.2. Packing and Stuffing Box

Packing materials shall be:

- PTFE-based for packing temperatures below 200°C.
- Graphite-based, metal-reinforced, for packing temperatures over 200 °C. Lubrication is not required, but for applications above 427 C° an extension or steel yoke should be used.

Packing shall not contain asbestos.

External lubricators or grease nipples shall not be applied.

Depending upon the design of the valve, an extended bonnet may be required to keep the temperature at the stuffing box at an acceptable value for the applied packing.

19.3. Gaskets

Body-to-bonnet and, if required, body-to-bottom flange gaskets shall be of the spiral wound type. Unless otherwise dictated by the process conditions, the gasket material shall be AISI 316 stainless steel, graphite filled, as a minimum.

19.4. Body Material

Valves body material shall in accordance with piping specification.

Cast iron bodies shall not be used.

For applications in sour gas service, not only body materials but also all metals exposed to line fluid shall be in compliance to section 5.8.

Bonnets shall be of the same material as the valve body and of integral or bolted type construction with fully retained gaskets. Threaded connections are not permitted.

19.5. Trim

Valve trim and seat materials shall be AISI 316 stainless steel as a minimum.

Valve trim materials shall be selected to suit the process conditions and fluid types.

Hardened trims shall be used where high differential pressure, pressure and temperature conditions.

Trims will be satellite for fluids containing solid particles or where vaporization or where sever wear may occur.

19.6. Bonnets

On globe style control valves, the bonnet shall be made the same as the valve body material or it is an equivalent forged material, as it is a pressure containing part subject to the line conditions.

Extension bonnets shall be used for either high or low temperature services to protect valve stem packing from extreme process temperatures.

19.7. Seat Leakage

Control valve seat leakage shall be designed and constructed in accordance with ANSI FCI 70.2.

Seat leakage class shall be follow ANSI/FCI B 16.104 or API 598

19.8. Yoke and Stem

Yokes shall be of suitable rigid material for open type construction and heavy duty.

Actuator stems shall have adequate strength to withstand maximum developed thrust of actuator.

All valves shall be equipped with a valve stem travel indicator.

19.9. Types of Actuators

General

Valve Actuators shall normally be spring return and diaphragm type. All actuators shall be adequate to stroke the valve under the maximum differential pressure to which the valve may be exposed.

The actuators shall have a position indicator.

Air operated diaphragms and springs shall be selected to optimize on a bench setting range of 0.2-1 Bar(g) for the specified maximum upstream pressure with the downstream pressure of zero bar. The "Bench Setting Range" and the "In Service Stroking Range" shall be specified on the control valve data plates.

The valve actuator shall be sized so the valve will operate with 10% more than the maximum indicated upstream pressure. The spring barrel shall accommodate interchangeable springs.

The Actuator spring shall be fully enclosed in a metal housing and permanently treated to resist atmospheric corrosion.

Pneumatic Spring Diaphragm Actuators

Pneumatic spring diaphragm actuators are the most frequently used device for positioning control valves and should be used whenever possible.

If operating conditions permit, ball valves should be equipped with long stroke diaphragm actuators instead of piston actuators.

Pneumatic Piston Actuators

Pneumatic piston actuators shall be used where requirement of high thrust long stroke, or higher speed of response and for other operating conditions that a diaphragm actuator is not suitable.

Piston operators shall be single acting spring-return type for throttling service, and these shall be used for special services or valve type.

Pistons and cylinders shall be of material suitable for withstanding the pressure and chemical characteristics of the operating medium over a wide range of ambient temperatures.

Electro Hydraulic Actuators

These actuators may only be used where the use of pneumatic actuators are not suitable or inappropriate.

Where used, hydraulic actuators shall be double acting cylinders mounted on the valves. The cylinders shall be constructed with stainless steel rod, screws and nuts as a minimum. Seals shall be viton as a minimum.

Actuator shall be totally enclosed and sealed to give protection to all internal moving parts.

The maximum torque needed for a valve stroke shall be multiplied by 1.3 safety coefficient at the minimum hydraulic pressure.

Unless otherwise specified, the maximum valve stroking time to achieve the safety position shall be 1 second per inch of the valve body size.

A stroke speed setting facility shall be made available and performed by means of needle valve restricting power actuating fluid.

The valve supply shall include all interconnecting piping between the hydraulic unit and the valve actuator. The interconnecting piping shall include all pipes and fittings, isolating valves, couplings, etc.

19.10. Control Valve Accessories

Smart Positioners

All modulating control valves shall be provided with Smart Positioners capable of handling HART signals. Non 'Hart' positioners may be used for special applications such as compressor Anti- Surge and will subject to Company agreement.

The valve positioners shall be mounted and piped to the valve yoke. Pneumatic positioners shall have a filter regulator, gauges and a bypass valve

The valve Positioners shall be sufficient capacity in both directions for pressuring and venting the actuator to prevent response time limitations.

Lock-up Valves

Air lock-up valves shall be provided for the following applications:

- All services requiring the control valve to remain in the position immediately prior to a complete failure of the instrument air supply.

- All shut-off control valves requiring an air supply pressure higher than the guaranteed minimum instrument air pressure.

The lock-up valves shall have a bolt adjustment provided with a locking facility to prevent tampering.

For control valves with a valve positioner, the lock-up shall be installed between the positioner output and the actuator. Where lock-up valves are applied on solenoid-operated valves, the solenoid valve shall be installed between the lock-up valve and the actuator.

Volume boosters

Volume boosters shall be provided if needed to achieve the stroking times specified in the requisition. Volume boosters for pneumatic actuators shall be of the high capacity type with fast throttling facilities to control the required capacity.

Limit Stops

Limit stops shall be fitted only if indicated on by the Licensor Design Recommendations.

Limit stops shall be mechanical devices mounted on the actuator. Screwed bolt- type limit stops, e.g. on the control valve stem, adjustable over the full length of the stroke shall be applied.

To prevent tampering, the limit stops shall be fitted with a locking facility, e.g. a locking nut.

The limit stops shall be adequately protected against unintentional adjustments.

Air filter regulator set

Each valve shall be supplied with a stainless steel filter regulator with purge and pressure gauge diam. 50mm fully tubed. Filter regulator connection shall be minimum 1/4" NPTF.

Limit switches

Valve limit switches shall be IS type inductive proximitors with volt free contacts. They shall be normally open, that means they shall be close where the position is reached.

Solenoid valves

Solenoid valves shall be fitted in airlines to the control valve.

Solenoid valves execution shall be Ex"d" 24VDC 2 wires low powered less than 4 W.

Solenoid valves shall be Ex d IIC T4-T6 (ATEX-Certified) as per IEC60079

Solenoid valve shall be energized during normal operation, de-energized on trip case.

The body material shall be 316 stainless steel, suitable for the environmental conditions and the service. Disc/seal material shall be resilient to give a tight shut off feature.

Air connection shall be 1/4" NPT, electric connection shall be M20. 1/2" NPT air connection may be considered regarding the time to fail to fail safe position.

Electrical or manual reset facility shall be provided as required.

Suitable dust/insect screens shall be provided on vents.

Tubing and fittings

Tubing & Fittings material shall be shall be 2507 duplex stainless steel.

Hand Wheel

Where specified, a hand wheel shall be side mounted and de-clutchable.

19.11. Control Valve Sizing

General Considerations

Control valves sizing shall be according to ISA S75.1.

The valve calculation sheet shall show the capacities, noise levels, and all the other related information.

Valves Sizing

Valves shall generally be selected to control with maximum limit operating conditions between 10% and 90% of its opening stroke. Control valves shall be sized such that the valve opening at normal flow condition to be around 60%-75% depending on trim characteristic.

Butterfly valves shall normally be sized for a maximum travel of 60 OC, unless the valve characteristics allow control over a wider range of opening.

19.12. Noise Level

The maximum noise level for each control valve shall be limited to 85 dBA for normal operation, measured at one meter away from the downstream pipework of the valve.

Where final noise calculation indicates more than 85 dB (A), higher scheduled pipes shall be considered for the downstream piping.

In those application where the use of low noise valves and increased line schedule still do not provide a low enough noise level, use of suitable acoustical insulation downstream and upstream of the valve will be required to meet the noise requirement.

19.13. Limitation in Outlet Velocity

The velocity in the valve outlet should not reach sonic velocity. Acoustic fatigue associated with large-flow gas piping systems shall be taken into consideration.

19.14. Special Valves

For Hydrogen services the control valves shall be variable Cv type with bellows seal follow piping spec to avoid a risk of leaks. The allowed seal leakage shall be in accordance with ANSI FCI 70.2.

20. ACTUATED VALVES

20.1. General

Valves shall generally be flanged, flange ANSI rating shall be in accordance with section 5.7

Valve body shall be cast carbon steel, at least to ASTM A216 grade WCB material, cast iron is not allowed.

Valve trim material (including the ball in ball valves) shall be 316 stainless steel, as a minimum. Higher grade materials shall be provided as required to meet the process condition.

For applications in sour gas services, as indicated in individual data sheets, material of body, trim and all other wetted metal parts in contact with the sour gas shall be in compliance to section 5.8

20.2. Blow Down Valves

Blow down Valves (BDV) is used for plant equipment depressurisation resulting from selected shutdown levels.

BDV's shall be initiated manually for process safeguarding, control and other operational reasons, including maintenance. The valve shall open fully when depressurisation is initiated.

BDV's shall have the following characteristics as a minimum:

- Low noise
- Air volume tank sized for three full open/close operations based on normal air supply pressure
- Tight shutoff (ANSI FCI 70.2, class VI)
- Fail to stay put actuator on polymer units. Shall be fail open for the others units

Valves shall be designed to open remotely from the control room. A 24 VDC signal shall be used to operate normally energised solenoid valve(s) to admit instrument air or hydraulic fluid to valve actuators. The valves shall return to their closed or opened position when the 24 VDC signal is removed.

BDV's shall be full-bore ball valves sized according to the required blow down flow rate and pressure drop.

Unless otherwise specified, BDV's shall be line size.

Open and close valve limit switches shall be provided where specified for on the Licensor Design Recommendations and these shall be fail safe IS type inductive proximity sensors with volt free contacts.

Actuators shall be sized to open and close the valves at the maximum differential pressures and within the time, as required by the process conditions.

Actuators shall be mounted on the valves.

20.3. ESD Valves

ESD valves are used for isolation of potentially hazardous flows of process fluids from one process system to another, as well as to the plant environment if containment is lost.

ESD valves shall have the following characteristics as a minimum:

- Actuator to close on air failure
- Fire safe design
- Tight shutoff (ANSI FCI 70.2, class VI)
- Remote reset facility.
- Fail to close
- 24 Volt DC Solenoid, de-energise to close

ESD valves shall be designed to open/close remotely from the control room through the ESD system, or as specified on Licensors Design Recommendations.

Open and close valve limit switches shall be provided where specified for on the P&ID and these shall be fail safe IS type inductive proximitors with volt free contacts.

Actuators shall be sized to fully open and close the valves at the maximum differential pressures and within the time spans as required by the process conditions.

In general, pneumatic, single acting, spring return actuators shall be used.

Actuators shall be mounted on the valves.

On-line testing facility shall be provided for ESD valves in accordance with IEC 61508 to meet the minimum 'testing interval'.

Seat leakage class shall be follow ANSI/FCI B 16.104 or API 598

20.4. Process Isolation Valves

Isolation valves shall be designed to open or close remotely from the control room through the ESD system. Failure position shall be stated on the Licensors Design Recommendations.

Actuators shall be sized to fully open and close the valves at the maximum differential pressures and within the time spans as required by the process conditions.

Isolation valves shall be full bore or reduced bore ball or plug valves according to the piping specification requirements.

Unless otherwise specified, Isolation valves shall be line size.

Open and close valve limit switches shall be provided where specified for on the Licensors Design Recommendations and these shall be fail safe IS type inductive proximitors with volt free contacts.

Actuators shall be pneumatic single action type, air to open, spring to close or air to close, spring to open.

20.5. Motor Operated Valves (MOV)

Motor operated valves are required for automatic isolation of some process equipment as indicated on the Licensors Design Recommendations.

**General Instrument
Standard**

Project No.:	01000
KUS Doc No.:	01000-IA-SPC-0001
Contr. Doc No.:	
Rev.:	R00
Page:	41 / 49

Unless otherwise specified, MOV's shall be line size.

Actuators shall be sized to open and close the valves at the maximum differential pressures and within the time, as required by the process conditions.

Actuators shall be mounted on the valves.

The actuator shall typically consist of the followings:

Three Phase Electric Motor

The motor shall be sufficiently sized to open and close the valve against maximum differential pressure when the voltage to motor terminal is within 10% of rated voltage of 400 V before starting. Unless otherwise specified, motor rating shall be 380 volts, 3-phase, 50 Hz.

The electric motor and accessories shall be EEx d type.

Limit Switches and Torque Switches

The actuator shall provide both torque and position limitation in both directions

Provision shall be made for the addition of extra sets of limit switches in each actuator. Each set shall be adjustable to any point of valve position. Each unit shall be provided with auxiliary contacts, rated at 1 Amp at 24 VDC for remote position indication.

Integral Control

The actuator shall have an integral motor starter, local control and lamp indication.

Local push buttons shall be provided for OPEN/CLOSE control of valve, with 3- position lockable selector switch providing Local/Off/Remote positions.

Local indication of CLOSE and OPEN positions shall be provided.

The actuator shall be fitted with microprocessor module forming part of the Motor operated control system as outlined in document Control Systems Philosophy and Specification 108184-0000-IA-SPC-0001.

The valves actuators shall be serially link and connected to MOV control system located at the appropriate Satellite Rack Room. Remote opening and closing of the MOV's shall be possible from the DCS, MOV control system and locally. Status of the valves shall also be transmitted via the serial link.

Hand wheel for Manual Operation

Hand wheels shall only be provided where shown on the Licensor Design Recommendations

Hand wheel shall close on Clockwise rotation of the actuator. The hand wheel drive shall be mechanically independent of the motor drive, and gearing shall allow for emergency manual operation in a reasonable time.

20.6. Deluge Valves

The deluge valves will be of the quick opening type, energised by firewater ring pressure. Preference is for straight through design to avoid possible obstruction during operation.

Valve size shall be limited to a maximum limit of 8"; the maximum pressure drop shall be limited to 0.7 bar(g).

Actuator shall be hydraulic single acting type, water to close, and spring to open.

The valves are normally kept closed under hydraulic (water) pressure. No external services shall be used to keep the valve closed; only the fire water main pressure is acceptable for this duty.

Automatically actuated valve shall open in the event of a fire. The means of achieving this shall be via a fusible plug system.

Remotely operated valve shall open via a command signal from the FGS system.

20.7. Pneumatic Type Actuators

General

The following actuators shall be selected for Emergency Shutdown, Blow Down and isolation valves

- Single acting piston type, spring return with spring to close action for air failure to close valves, and spring to open on air failure to open valves
- Double acting piston type shall be avoided.

Actuators provided for ESD Valves shall be equipped with on-line testing facilities.

Short stroke single acting pneumatic piston actuators shall preferably be used on emergency shutdown services.

A stroke speed setting facility in both directions shall be made available and performed by means of needle valve restricting power actuating fluid.

Accessories such as solenoid valves, limit switches, etc., shall be installed directly on the actuator or on the body of the valve in order to minimise the valve overall time response.

Air Volume Tank

Generally BDV Actuators and any double acting actuators shall be equipped with an air tank manufactured from carbon steel and calculated as per ASME VIII.

The capacity shall be sized to permit 3 full valve strokes, where one stroke is open and close, in case of air failure.

Small sized air volume tanks shall be mounted directly on the valve itself. Other tanks shall be free standing and separate from the valves.

Accessories

Accessories such as solenoid valves, electrical position switches, pressure pilots, pneumatic relays, etc. shall be supplied and mounted by the valve Vendor.

- Air filter regulator set - Each valve shall be supplied with a stainless steel filter regulator and pressure gauge fully tubed.
- Limit switches - Valve limit switches shall be fail-safe IS type inductive proximitors with volt free contacts.
- Solenoid valves - The body material shall be 316 stainless steel, suitable for the environmental conditions and the service. Disc/seat material shall be resilient to give a tight shut off feature.
- Electrical or manual reset facility shall be provided as required.
- Tubing and fittings - Tubing & Fittings material shall be shall be 2507 duplex stainless steel.
- Hand wheel - Where specified, a hand wheel shall be side mounted and de-clutchable.
- Position indicator - On/off actuators shall be provided with mechanical position indicator devices matching the two positions of the valve.

These components and all other necessary facilities for the operation of the actuators shall form part of one assembly, completely tubed, wired and tested. The assembly shall be housed in a weatherproof AISI 316 housing suitable for the specified hazardous area, and shall only require the connection of instrument air supply and wiring of solenoid valves and limit switches. All tubing for pneumatic circuits shall be 2507 duplex stainless steel. as minimum. Junction boxes shall be stainless steel.

20.8. Hydraulic Type Actuators

The hydraulic actuators where required will be double acting cylinders mounted on the valves. The cylinders shall be constructed with stainless steel rod, screws and nuts as a minimum. Seals shall be viton as a minimum.

Hydraulic accumulators shall be of the piston type, nitrogen charged, with piston position detection possibility. The accumulator shall be installed close to the valve.

Actuator shall be totally enclosed and sealed to give protection to all internal moving parts.

The minimum hydraulic pressure shall be at least 1.3 times the maximum torque needed for a valve stroke.

Unless otherwise specified, the maximum valve stroking time to achieve the safety position shall be 1 second per inch of the valve body size.

A stroke speed setting facility in both directions shall be made available and performed by means of needle valve, restricting power-actuating fluid.

Components shall all be AISI 316.

21. SAFETY RELIEF VAVES

Refer to Pressure Relief Valves Standard 108184-0000-PR-SPC-0010

22. FIRE AND GAS DETECTION

22.1. General

Fire and Gas detector shall report to the Fire and Gas System. The Fire and Gas detector shall follow the general requirements of the instruments.

All the Gas detectors and Manual Call Point shall be suitable for use in zone I. For the fire detection of buildings, an independent addressable fire detection system is foreseen. The associated smoke detectors and MAC's will be connected on an addressable loop to the central processing unit.

Contractor shall define the technology to be employed, the quantities to provide, the location for each fire and gas detector according to the HSE Design Criteria.

The following type of detector may be used:

- Flammable gas detector (IR)
- Toxic gas (H₂S)
- Manual Call Point
- Smoke detector (Optical and ionisation)
- Very Early Smoke Detector Alarm
- Heat detector
- Addressable fire and smoke detectors (inside building)

22.2. Fire Extinguishing System

A fire extinguishing system shall be dedicated to fire protection. This system shall include at minimum the following equipments:

- Fire pumps
- Deluge valves
- Loops with fusible plugs
- Hydrants
- Foam tank
- Sprinklers

23. PACKAGE EQUIPMENT

Instrumentation for package equipment shall be furnished as a complete package. All instruments in skid-mounted package shall be supplied fully piped and wired. All wiring or pneumatic tubing for remote transmission shall be terminated at the edge of the skid in junction boxes/ instrument air manifold or blind flange.

Note: pneumatic signals refer to valves actuators only.

All instrumentation types make &model, wiring, installation, junction boxes, connection to piping and equipment, etc. shall identical to main plant instrumentation. It is expected that the above stated requirements may not be possible for some specialised packages such mechanical handling complex turbo- machinery, any exception to the above instrumentation requirements shall be reviewed on a case by case basis and agreed to with Company.

Instruments shall uniquely numbered and in accordance with the project numbering specification. Instruments symbols shall be in accordance with the instruments symbols rules.

INtools instruments data base shall be implemented for all package instruments and associated cabling wiring, etc. Contractor shall be responsible for consolidating the overall INtools instruments data base across the entire complex.

For implementation of control, monitoring and safety systems for the package equipments refer to Control Systems Philosophy and Specification and PACKAGE UNIT INSTRUMENTATION STANDARD (to be provided by Contractor).

24. ANALYSERS AND SAMPLING SYSTEMS

24.1. Analysers

Analysers shall be provide as required by the Licensor Design Recommendations where they are required to ensure product quality/purity, or where required by environmental and safety regulation.

The following on line analyser types, as a minimum, shall be considered by Contractor;

- Gas chromatographs
- Hydrocarbons analysers
- Moisture analysers
- Oxygen analysers
- SO_x, NO_x, CO
- Density analysers
- Conductivity and PH analysers
- Etc.

All analyser's equipments shall be microprocessor based providing high degree of self diagnostic.

Analysers and analyser houses shall be designed according to IEC 61285, IEC 60079-16, IEC 61115 and local applicable codes and standards.

Analysers required for process control and safety duty shall be hardwired with 4-20mA directly to the DCS/ ESD.

For gas chromatographs redundant serial links to the DCS can be utilized depending on the numbers of GCs and the associated data required for control and displays.

Analysers shall be located in Analyser Houses with appropriate heating, ventilation and air conditioning systems should be outlined in Doc. Control Systems Philosophy and Specification (to be developed by Contractor). Analysers shall be selected in accordance with their required process analysis.

The analysers shall be provided with Analyser Management System (AMS) used for maintenance and data acquisition of the analysers to provide automatic data collection, logging the performance of the analysers, validation of calibration data, statistical data handling, etc. as outlined in the above mentioned philosophy document.

24.2. Sampling Systems

The analyser sample systems need to extract, transport and condition the sample at the analyser at the analyser inlet as required by the process analysis requirement. The design shall ensure that none of the sample properties are altered between samples take-off point and analyser inlet.

The system shall be designed to provide ease of operation and maintenance.

Each analyser shall have its own sample delivery and conditioning system.

In general, the analyser sample systems shall be installed in enclosure and mounted outside the analyser houses under a canopy for weather protection.

Calibration gases bottles shall also installed outside the analyser houses under the canopy.

Moisture analyser will have field mounted sample system with sample transport line as short as possible.

The optimum installation configuration must be determined on a case by case basis to suit the required application.

Typically the complete sample system will consists of the following components:

- Sample tap with sample probe (sample take-off)
- Sample pre-conditioning
- Sample transport lines
- Sample conditioning with bypass and isolation valves.
- Analyser inlet with calibration inlet stream switching
- Calibration gases
- Effluent return and /or vent

Tubing and fittings for the analysers and sampling systems will be of duplex 2507 (or 2205) seamless stainless steel precision types as minimum. For corrosive services high grade materials such as Monel, Hastelloy, etc shall be used.

Sizes of tubing of 1/4, 3/8 and 1/2 inch are the preferred sizes.

Fittings for connecting the tubing will be by means of double ferrule compression ring type 2507 (or 2205) sst. (Swagelok, Gyrolok or equivalent-quality) compression fittings.

PTFE tubing where used will be installed and connected with the aid of special thread- type fittings.

24.3. Analyser Houses

Analyser houses shall be provided across the complexes and located as appropriately in a process unit or process area. Location of the houses shall be such that the sample lines distances be as short as possible between the sample take points and the analyser in order to minimise the transportation time of the analysers samples.

The analyser houses shall take in account all weather conditions, and shall be constructed from stainless steel frames with roof and wall cladding construction mounted on a fabricated steel beam frame.

The analyser house roof overhang shall provide weather protection for the maintenance of the sample conditioning systems. The roof overhang shall be 1000 mm on all and fire rating.

In addition to mounting and installing the analysers, the sample conditioning system, tubing, fittings, cabling, etc. the Analyser Houses shall be provided as required with:

- A common header for atmospheric venting, installed overhead
- A common drain header connected to the plant drain system
- A common Instrument air header and distribution system
- HVAC system complete with air intake ducting and ancillary equipment
- Calibration gas cylinder racks complete regulators and accessories
- Carrier gas manifold station and auto-change over system with associated pressure regulators and gauges.
- A common Nitrogen header
- Internal and external lighting
- Drinking water facilities sink and taps
- A free standing shelter for the carrier gas manifold bay.
- Etc.

The complete Analyser House, analysers, sampling systems all analyser ancillary equipment shall be assembled and the complete systems tested prior to transportation to site as a single unit where possible .sides of the analyser house.

All Analyser House walls penetrations for cables, pipes and instruments tubing shall be through separate transit frames and made of modular construction. The transit frame system shall maintain the Analyser House design criteria for gas tight seal.

25. PAINTING

As general rule, all the in-line portions of instruments shall meet the painting system regarding the piping specifications Standard for Protective Coatings and Application Requirements 108184-0000-MA-SPC-0009

All the others portions like actuators, housings, transmitters, etc vendor's painting.

Painting system shall take into account the local climatic conditions; refer to paragraph 5.1 Environmental conditions.

26. INSPECTION AND TESTING

Requirements of non-destructive testing like, radiography, magnetic particle test, hardness test, hydro test for items such as control valves, safety relief valves, level gauges etc.. shall be carried out as per respective piping classes.

For all instruments, welding and testing shall be carried out as per piping specifications. Post weld heat treatment shall be carried out as required by the piping material specifications.

Following test shall be carried out at vendor's works and test certificate shall be furnished.

- Material test certificate including NACE requirement for all instruments.
- Visual test.
- Calibration test certificates for all instruments including control valves, On/Off valves, safety relief valves.
- Seat leakage test for control valves, On/Off valves, Safety relief valves.
- Helium leakage test for control valves, On/Off valves, Safety relief valves in H2S service.
- Test certificate for safety relief valves set pressure.
- Hydro test for all in-line instruments, pipes, tubes, etc...
- Radiography of control valves, On/Off valves, safety relief valves of rating 600 lbs and above.
- NACE compliance certificate..
- Cryogenic test for all in cryogenic device.

27. SPARE PARTS

As a general rule, spare parts are required for all type of equipments and shall be in accordance with the project document

Contractor shall clearly list required spare parts and select them in order to minimise the amount of model numbers.

28. DOCUMENTATION

The official language shall be English. All text documents shall be generated from Windows Word, Excel latest release.

The Russian and/or Kazakh language shall be used in operation manual and human manual interface.

Contractor shall provide a list of document fully completed including but not limited to the following documents:

- Planning
- General and particular specifications, design documents
- Calculation notes
- Certifications, procedures
- Instruments, cables, Material Take Off schedule
- Loops diagram, functional and logic diagrams
- Drawings like cable routing, instruments and junction boxes location, F&G location, building arrangement, hook up, wiring, interconnections, etc...
- Vendor list, subcontractor list
- Detail studies like local panel, support boo
- Spare parts list
- As built documents